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Last login: Tue Jun 16 13:45:13 on ttys002
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch
psql (18.4)
Type "help" for help.

tpch=# -- Step 1: Reset role and disable RLS
RESET ROLE;
ALTER TABLE orders DISABLE ROW LEVEL SECURITY;

-- Step 2: Drop existing view if any
DROP VIEW IF EXISTS orders_secure;

-- Step 3: Create view with same filter logic as RLS policy
CREATE VIEW orders_secure AS
SELECT o.*
FROM orders o
WHERE EXISTS (
    SELECT 1 FROM customer c
    WHERE c.c_custkey = o.o_custkey
    AND c.c_acctbal > 5000
);

-- Step 4: Grant permissions and set role
GRANT SELECT ON orders_secure TO alice;
GRANT SELECT ON customer TO alice;
GRANT SELECT ON lineitem TO alice;
SET ROLE alice;
RESET
ALTER TABLE
DROP VIEW
CREATE VIEW
GRANT
GRANT
GRANT
SET
tpch=> select
    o_orderpriority,
    count(*) as order_count
from
    orders_secure
where
    o_orderdate >= date '1994-01-01'
    and o_orderdate < date '1994-01-01' + interval '3' month
    and exists (
        select
            *
        from
            lineitem
        where
            l_orderkey = o_orderkey
            and l_commitdate < l_receiptdate
    )
group by
    o_orderpriority
order by
    o_orderpriority;
WARNING: Logix Planner Input:
DETAIL:  {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
:alias <>
:eref
  {ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 1
:subquery
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
:alias
  {ALIAS
:aliasname o
:colnames <>
}
}
:eref
  {ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtekind 0
:relid 16561
:inh true
:relkind r
:rellockmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
(FROMEXPR
:fromlist (
  {RANGETBLREF
:rtindex 1
}
)
)

```

```

:quals
{SUBLINK
:subLinkType 0
:subLinkId 0
:testexpr <>
:operName <>
:subselect
{QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
{RANGETBLENTY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationk
ey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh true
:relkind r
:rellocKmode 1
:perminfoindex 1
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointrree
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 1
}
)
}
:quals
{BOOLEXP
:boolop and
:args (
{OPEXP
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 1
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
}
)
:location -1
}
{OPEXP
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 1
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
{FUNCEXP
:funcid 316
:funcresulttype 701
:funcset false
:funcvariadic false
:funcformat 2
:funccollid 0
:inputcollid 0
:args (
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ -120 19 0 0 0 0 0 0 ]

```

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        }
        }
        :location -1
    }
    }
    :location -1
}
}
:location -1
}
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{CONST
:consttype 23
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 1 0 0 0 0 0 0 0 ]
}
:resno 1
:resname ?column?
:ressortgrouppref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderkey
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location -1
}
:resno 2
:resname o_custkey
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 160
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location -1
}
:resno 3
:resname o_orderstatus
:ressortgrouppref 0
:resorigtbl 16561
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:vartype 781
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location -1
}
:resno 4
:resname o_totalprice
:ressortgrouppref 0

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:resorigtbl 16561
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varatype 1882
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname o_orderdate
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
:resno 6
:resname o_orderpriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
:resno 7
:resname o_clerk
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
:resno 8
:resname o_shippriority
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:varatype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
:resno 9
:resname o_comment
:ressortgroupref 0
:resorigtbl 16561
:resorigcol 9
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 16973
:inh false
:relkind v
:relockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTRY

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:alias <>
:eref
{
  ALIAS
  :aliasname *GROUP*
  :colnames ("*o_orderpriority")
}
:rtekind 9
:groupexprs (
  {VAR
    :varno 1
    :varattno 6
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 6
    :location 11
  }
)
:lateral false
:inFromCl false
:securityQuals <>
}
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 16973
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 12 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
)
:quals
{BOOLEXPRESS
  :boolop and
  :args (
    {OPEXPRESS
      :opno 1098
      :opfuncid 1090
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 5
          :vartype 1082
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 5
          :location 89
        }
        {CONST
          :consttype 1082
          :consttypmod -1
          :constcollid 0
          :constlen 4
          :constbyval true
          :constisnull false
          :location 109
          :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
        }
      )
    }
    :location 101
  )
  {OPEXPRESS
    :opno 2345
    :opfuncid 2338
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 5
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 5
        :location 130
      }
      {OPEXPRESS
        :opno 1076
        :opfuncid 2071
        :opresulttype 1114
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {CONST
            :consttype 1082
            :consttypmod -1
            :constcollid 0
            :constlen 4
            :constbyval true
            :constisnull false
            :location 149
            :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
          }
          {CONST
            :consttype 1186
            :consttypmod 196607
            :constcollid 0
            :constlen 16
            :constbyval false
            :constisnull false
            :location 173
            :constvalue 16 [ 0 0 0 0 0 0 0 0 0 0 0 0 0 3 0 0 0 ]
          }
        )
      }
    }
    :location 162
  )
}
:location 142
}
{SUBLINK
  :subLinkType 0
  :subLinkId 0
  :testexpr <>
  :operName <>
  :subselect
  {QUERY
    :commandType 1
    :querySource 0
    :canSetTag true
  }
}

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:utilityStmt <>
:resultRelation 0
:hasAggs false
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTRY
    :alias <>
    :eref
    {ALIAS
      :aliasname lineitem
      :colNames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenu
mber" "l_quantity" "l_extendedprice" "l_discount" "l_tax"
"l_returnflag" "l_linestatus" "l_shipdate" "l_commitdate
" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_commen
t")
    }
    :rtekind 0
    :relid 16566
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rtepermInfos (
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22
23)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
}
:quals
{BOOLEXP
  :boolop and
  :args (
    {OPEXP
      :opno 96
      :opfuncid 65
      :opresulttype 16
      :opretset false
      :opcollid 0
      :inputcollid 0
      :args (
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location 289
        }
        {VAR
          :varno 1
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 1
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 1
          :location 302
        }
      )
    }
    :location 300
  }
  {OPEXP
    :opno 1095
    :opfuncid 1087
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 1
        :varattno 12
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 12
        :location 329
      }
      {VAR
        :varno 1
        :varattno 13
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 1
        :varattnosyn 13
        :location 344
      }
    )
  }
  :location 342
}
:location 325
}
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
    }
  }
)

```

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:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 1
:location 227
}
}
:resno 1
:resname 1_orderkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 1
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 2
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 2
:location 227
}
}
:resno 2
:resname 1_partkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 2
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 3
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 3
:location 227
}
}
:resno 3
:resname 1_suppkey
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 3
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 4
:varstype 23
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 4
:location 227
}
}
:resno 4
:resname 1_linenummer
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 4
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 5
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location 227
}
}
:resno 5
:resname 1_quantity
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location 227
}
}
:resno 6
:resname 1_extendedprice
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:varstype 701
:varstypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location 227
}
}
:resno 7
:resname 1_discount
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 7
:resjunk false

```

```

}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location 227
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location 227
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location 227
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location 227
}
:resno 11
:resname l_shipdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location 227
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location 227
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location 227
}
}

```

```

:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location 227
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location 227
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 16566
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}
:location 191
}
:location 126
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 2
:varattnosyn 1
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16973
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 32
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returninglist <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664

```

```

:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
  {SORTGROUPCLAUSE
   :tieSortGroupRef 1
   :eqop 98
   :sortop 664
   :reverse_sort false
   :nulls_first false
   :hashable true
  }
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 421
}

WARNING: Log 2 : After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE true
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
   :alias <>
   :eref
   {ALIAS
    :aliasname orders_secure
    :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
   }
   :rtekind 1
   :subquery <>
   :security_barrier false
   :relid 16973
   :inh false
   :relkind v
   :rellockmode 1
   :perminfoindex 1
   :lateral false
   :inFromCl true
   :securityQuals <>
  }
  {RANGETBLENTY
   :alias <>
   :eref
   {ALIAS
    :aliasname *GROUP*
    :colnames ("o_orderpriority")
   }
   :rtekind 9
   :groupexprs (
    {VAR
     :varno 4
     :varattno 6
     :vartype 25
     :vartypmod -1
     :varcollid 100
     :varnullingrels (b)
     :varlevelsup 0
     :varreturningtype 0
     :varnosyn 4
     :varattnosyn 6
     :location -1
    }
   )
   :lateral false
   :inFromCl false
   :securityQuals <>
  }
  {RANGETBLENTY
   :alias <>
   :eref
   {ALIAS
    :aliasname lineitem
    :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenummer" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
   }
   :rtekind 0
   :relid 16566
   :inh false
   :relkind r
   :rellockmode 1
   :perminfoindex 2
   :tablesample <>
   :lateral false
   :inFromCl true
   :securityQuals <>
  }
  {RANGETBLENTY
   :alias
   {ALIAS
    :aliasname o
    :colnames <>
   }
   :eref
   {ALIAS
    :aliasname o
    :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
   }
   :rtekind 0
   :relid 16561
   :inh false
   :relkind r
   :rellockmode 1
   :perminfoindex 3
   :tablesample <>
   :lateral false
   :inFromCl true
   :securityQuals <>
  }
  {RANGETBLENTY
   :alias
   {ALIAS
    :aliasname c
    :colnames <>
   }
  }
)
}

```

```

:eref
{
  ALIAS
  :aliasname c
  :colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "
    c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 16973
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
)
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (
{RANGETBLREF
:rtindex 4
}
)
:quals <>
}
):rarg
{RANGETBLREF
:rtindex 5
}
):usingClause <>
:join_using_alias <>
:quals (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
)
):location -1
}
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true

```

```

        :constisnull false
        :location -1
        :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
    }
    :location -1
}
)
:alias <>
:rtindex 0
}
)
:qualis (
{OPEXPR
:opno 1098
:opfuncid 1090
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
){CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location 109
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location 101
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
){CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
:location 142
}
}
)
}
:arg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:qualis (
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 289
}
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
)
:location 300
}
{OPEXPR
:opno 1095
:opfuncid 1007
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location 329
}

```

```

        {VAR
        :varno 3
        :varattno 13
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 13
        :location 344
        }
    )
    :location 342
    }
    )
    :alias <>
    :rtindex 0
    }
    )
    :quals <>
    }
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 1
:resname o_orderpriority
:ressortgroupref 1
:resorigtbl 16973
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{AGGREF
:aggfnoid 2803
:aggtype 20
:aggcollid 0
:inputcollid 0
:aggtranstype 0
:aggargtypes <>
:aggdirectargs <>
:args <>
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar true
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 32
}
:resno 2
:resname order_count
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause (
{SORTGROUPCLAUSE
:tleSortGroupRef 1
:eqop 98
:sortop 664
:reverse_sort false
:nulls_first false
:hashable true
}
)
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 421
}

```

WARNING: Log 3: At the starting of Plan creation:

```

DETAIL: {AGGPATH
:path.pathtype 365
:parent_relds (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 5
:path.disabled_nodes 0
:path.startup_cost 194594.51589866364
:path.total_cost 194617.5859964136
:path.pathkeys (
{PATHKEY
:pk_eclass
{EQUIVALENCECLASS
:ec_ofamilies (o 1994 2095)
:ec_collation 100
:ec_childmembers_size 0
:ec_members (
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 4
:varattno 6
:vartype 25

```

```

        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 6
        :location -1
    }
    :em_relids (b 4)
    :em_is_const false
    :em_is_child false
    :em_datatype 25
    :em_jdomain
    {JOINDOMAIN
    :jd_relids (b 3 4 5)
    }
    }
    :ec_childmembers <>
    :ec_sources <>
    :ec_derives_list <>
    :ec_relids (b 4)
    :ec_has_const false
    :ec_has_volatile false
    :ec_broken false
    :ec_sortref 1
    :ec_min_security 4294967295
    :ec_max_security 0
    }
    :pk_opfamily 1994
    :pk_cmptype 1
    :pk_nulls_first false
    }
)
:subpath
{GATHERMERGEPATH
:path.pathtype 369
:parent_relids (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe false
:path.parallel_workers 0
:path.rows 12
:path.disabled_nodes 0
:path.startup_cost 194594.51589866364
:path.total_cost 194617.47599641362
:path.pathkeys (
{PATHKEY
:pk_eclass
{EQUIVALENCECLASS
:ec_opfamilies (o 1994 2095)
:ec_collation 100
:ec_childmembers_size 0
:ec_members (
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:em_relids (b 4)
:em_is_const false
:em_is_child false
:em_datatype 25
:em_jdomain
{JOINDOMAIN
:jd_relids (b 3 4 5)
}
}
)
:ec_childmembers <>
:ec_sources <>
:ec_derives_list <>
:ec_relids (b 4)
:ec_has_const false
:ec_has_volatile false
:ec_broken false
:ec_sortref 1
:ec_min_security 4294967295
:ec_max_security 0
}
:pk_opfamily 1994
:pk_cmptype 1
:pk_nulls_first false
}
)
:subpath
{AGGPATH
:path.pathtype 365
:parent_relids (b)
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe true
:path.parallel_workers 2
:path.rows 5
:path.disabled_nodes 0
:path.startup_cost 193594.49212422612
:path.total_cost 193616.06712422607
:path.pathkeys (
{PATHKEY
:pk_eclass
{EQUIVALENCECLASS
:ec_opfamilies (o 1994 2095)
:ec_collation 100
:ec_childmembers_size 0
:ec_members (
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:em_relids (b 4)
:em_is_const false
:em_is_child false
:em_datatype 25
:em_jdomain
{JOINDOMAIN
:jd_relids (b 3 4 5)
}
}
)
:ec_childmembers <>
:ec_sources <>
:ec_derives_list <>
:ec_relids (b 4)
:ec_has_const false
:ec_has_volatile false
:ec_broken false

```

```

:ec_sortref 1
:ec_min_security 4294967295
:ec_max_security 0
}
:pk_opfamily 1994
:pk_cmptype 1
:pk_nulls_first false
}
)
:subpath
{SORTPATH
:path.pathtype 362
:parent_relids (b)
:path.pathtarget
{PATHTARGET
:exprs (
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
)
:sortgrouprefs ( 1)
:cost.startup 0
:cost.per_tuple 0
:width 9
:has_volatile_expr 0
}
:required_outer (b)
:path.parallel_aware false
:path.parallel_safe true
:path.parallel_workers 2
:path.rows 2870
:path.disabled_nodes 0
:path.startup_cost 193594.49212422612
:path.total_cost 193601.6671242261
:path.pathkeys (
{PATHKEY
:pk_eclass
{EQUIVALENCECLASS
:ec_optfamilies (o 1994 2095)
:ec_collation 100
:ec_childmembers_size 0
:ec_members (
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
}
:em_relids (b 4)
:em_is_const false
:em_is_child false
:em_datatype 25
:em_jdomain
{JOINDOMAIN
:jd_relids (b 3 4 5)
}
}
)
:ec_childmembers <>
:ec_sources <>
:ec_derives_list <>
:ec_relids (b 4)
:ec_has_const false
:ec_has_volatile false
:ec_broken false
:ec_sortref 1
:ec_min_security 4294967295
:ec_max_security 0
}
:pk_opfamily 1994
:pk_cmptype 1
:pk_nulls_first false
}
)
:subpath
{HASHPATH
:jpath.path.pathtype 359
:parent_relids (b 3 4 5)
:jpath.path.pathtarget
{PATHTARGET
:exprs (
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
)
:sortgrouprefs ( 1)
:cost.startup 0
:cost.per_tuple 0
:width 9
:has_volatile_expr 0
}
:required_outer (b)
:jpath.path.parallel_aware true
:jpath.path.parallel_safe true
:jpath.path.parallel_workers 2
:jpath.path.rows 2870
:jpath.path.disabled_nodes 0
:jpath.path.startup_cost 154455.46666666667
:jpath.path.total_cost 193429.6560416667
:jpath.path.pathkeys <>
:jpath.jointype 4
:jpath.inner_unique false
:jpath.outerjoinpath
{HASHPATH
:jpath.path.pathtype 359
:parent_relids (b 4 5)
:required_outer (b)
:jpath.path.parallel_aware true
:jpath.path.parallel_safe true
:jpath.path.parallel_workers 2
:jpath.path.rows 10210
:jpath.path.disabled_nodes 0
:jpath.path.startup_cost 4806.4125
:jpath.path.total_cost 40362.284374999996
:jpath.path.pathkeys <>
:jpath.jointype 4
:jpath.inner_unique false
:jpath.outerjoinpath
{PATH

```

```

:pathtype 339
:parent_relids (b 4)
:required_outer (b)
:parallel_aware true
:parallel_safe true
:parallel_workers 2
:rows 22585
:disabled_nodes 0
:startup_cost 0
:total_cost 35383
:pathkeys <>
}
:jpath.innerjoinpath
{PATH
:pathtype 339
:parent_relids (b 5)
:required_outer (b)
:parallel_aware true
:parallel_safe true
:parallel_workers 2
:rows 28253
:disabled_nodes 0
:startup_cost 0
:total_cost 4453.25
:pathkeys <>
}
:jpath.joinrestrictinfo (
{RESTRICTINFO
:clause
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
}
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
}
:location -1
}
:is_pushed_down true
:can_join true
:pseudocostant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 2
:security_level 0
:num_base_rels 2
:clause_relids (b 4 5)
:required_relids (b 4 5)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 4)
:right_relids (b 5)
:orclause <>
:rinfo_serial 7
:reval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 6.666666666666667e-06
:outer_selec 0.4520533333333333
:mergeopfamilies (o 1976)
:left_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
:em_relids (b 4)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relids (b 3 4 5)
}
}
:right_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:em_relids (b 5)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relids (b 3 4 5)
}
}
:outer_is_left false
:hashjoinoperator 96
:left_bucketsize 0.0003252032520325203
:right_bucketsize 1.4747522416234073e-05
:left_mcvfreq 0
:right_mcvfreq 0
:left_hasheqoperator 96
:right_hasheqoperator 96
}
)

```

```

:path_hashclauses (
  {RESTRICTINFO
    :clause
      {OPEXPR
        :opno 96
        :opfuncid 65
        :opresulttype 16
        :opretset false
        :opcollid 0
        :inputcollid 0
        :args (
          {VAR
            :varno 4
            :varattno 2
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 4
            :varattnosyn 2
            :location -1
          }
          {VAR
            :varno 5
            :varattno 1
            :vartype 23
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 5
            :varattnosyn 1
            :location -1
          }
        )
      }
    :location -1
  }
  :is_pushed_down true
  :can_join true
  :pseudoconstant false
  :has_clone false
  :is_clone false
  :leakproof false
  :has_volatile 2
  :security_level 0
  :num_base_rels 2
  :clause_relids (b 4 5)
  :required_relids (b 4 5)
  :incompatible_relids (b)
  :outer_relids (b)
  :left_relids (b 4)
  :right_relids (b 5)
  :orclause <>
  :rinfo_serial 7
  :eval_cost.startup 0
  :eval_cost.per_tuple 0.0025
  :norm_selec 6.666666666666667e-06
  :outer_selec 0.4520533333333333
  :mergeopfamilies (o 1976)
  :left_em
    {EQUIVALENCEMEMBER
      :em_expr
        {VAR
          :varno 4
          :varattno 2
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 4
          :varattnosyn 2
          :location -1
        }
        :em_relids (b 4)
        :em_is_const false
        :em_is_child false
        :em_datatype 23
        :em_jdomain
          {JOINDOMAIN
            :jd_relids (b 3 4 5)
          }
        }
      :right_em
        {EQUIVALENCEMEMBER
          :em_expr
            {VAR
              :varno 5
              :varattno 1
              :vartype 23
              :vartypmod -1
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
              :varreturningtype 0
              :varnosyn 5
              :varattnosyn 1
              :location -1
            }
            :em_relids (b 5)
            :em_is_const false
            :em_is_child false
            :em_datatype 23
            :em_jdomain
              {JOINDOMAIN
                :jd_relids (b 3 4 5)
              }
            }
          :outer_is_left false
          :hashjoinoperator 96
          :left_bucketsize 0.0003252032520325203
          :right_bucketsize 1.4747522416234073e-05
          :left_mcvfreq 0
          :right_mcvfreq 0
          :left_hasheqoperator 96
          :right_hasheqoperator 96
        }
      }
    :num_batches 1
    :inner_rows_total 67807.2
  }
  :jpath.innerjoinpath
    {PATH
      :pathtype 339
      :parent_relids (b 3)
      :required_outer (b)
      :parallel_aware true
      :parallel_safe true
      :parallel_workers 2
      :rows 833461
      :disabled_nodes 0
      :startup_cost 0
      :total_cost 135974.79166666666
      :pathkeys <>
    }
  :jpath.joinrestrictinfo (
    {RESTRICTINFO
      :clause
        {OPEXPR
          :opno 96
          :opfuncid 65
          :opresulttype 16
          :opretset false
          :opcollid 0

```

```

:inputcollid 0
:args (
  {VAR
    :varno 4
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 1
    :location -1
  }
  {VAR
    :varno 3
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 3
    :varattnosyn 1
    :location 289
  }
)
:location -1
}
:is_pushed_down true
:can_join true
:pseudoconstant false
:has_clone false
:is_clone false
:leakproof false
:has_volatile 2
:security_level 0
:num_base_rels 2
:clause_relids (b 3 4)
:required_relids (b 3 4)
:incompatible_relids (b)
:outer_relids (b)
:left_relids (b 4)
:right_relids (b 3)
:orclause <>
:rinfo_serial 8
:eval_cost.startup 0
:eval_cost.per_tuple 0.0025
:norm_selec 6.666666666666667e-07
:outer_selec 0.28109466666666666
:mergeopfamilies (o 1976)
:left_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
  :varno 4
  :varattno 1
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 4
  :varattnosyn 1
  :location -1
}
:em_relids (b 4)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relids (b 3 4 5)
}
}
:right_em
{EQUIVALENCEMEMBER
:em_expr
{VAR
  :varno 3
  :varattno 1
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 3
  :varattnosyn 1
  :location 289
}
:em_relids (b 3)
:em_is_const false
:em_is_child false
:em_datatype 23
:em_jdomain
{JOINDOMAIN
:jd_relids (b 3 4 5)
}
}
:outer_is_left false
:hashjoinoperator 96
:left_bucketsize 1.8449163330442964e-05
:right_bucketsize 7.115857596391243e-06
:left_mcvfreq 0
:right_mcvfreq 0
:left_hasheqoperator 96
:right_hasheqoperator 96
}
)
:path_hashclauses (
{RESTRICTINFO
:clause
{OPEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
  :varno 4
  :varattno 1
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 4
  :varattnosyn 1
  :location -1
}
{VAR
  :varno 3
  :varattno 1
  :vartype 23
  :vartypmod -1
  :varcollid 0
  :varnullingrels (b)
  :varlevelsup 0
  :varreturningtype 0
  :varnosyn 3
  :varattnosyn 1

```

```

        :location 289
      }
    }
    :location -1
  }
  :is_pushed_down true
  :can_join true
  :pseudoconstant false
  :has_clone false
  :is_clone false
  :leakproof false
  :has_volatile 2
  :security_level 0
  :num_base_rels 2
  :clause_relds (b 3 4)
  :required_relds (b 3 4)
  :incompatible_relds (b)
  :outer_relds (b)
  :left_relds (b 4)
  :right_relds (b 3)
  :orclause <>
  :rinfo_serial 8
  :eval_cost.startup 0
  :eval_cost.per_tuple 0.0025
  :norm_selec 0.6666666666666667e-07
  :outer_selec 0.28107466666666666
  :mergeopfamilies (o 1976)
  :left_em
    {EQUIVALENCEMEMBER
      :em_expr
        {VAR
          :varno 4
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 4
          :varattnosyn 1
          :location -1
        }
        :em_relds (b 4)
        :em_is_const false
        :em_is_child false
        :em_datatype 23
        :em_jdomain
          {JOINDOMAIN
            :jd_relds (b 3 4 5)
          }
      }
    }
  :right_em
    {EQUIVALENCEMEMBER
      :em_expr
        {VAR
          :varno 3
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 3
          :varattnosyn 1
          :location 289
        }
        :em_relds (b 3)
        :em_is_const false
        :em_is_child false
        :em_datatype 23
        :em_jdomain
          {JOINDOMAIN
            :jd_relds (b 3 4 5)
          }
      }
    }
  :outer_is_left false
  :hashjoinoperator 96
  :left_bucketsize 1.8449163330442964e-05
  :right_bucketsize 7.115857596391243e-06
  :left_mcvfreq 0
  :right_mcvfreq 0
  :left_hasheqoperator 96
  :right_hasheqoperator 96
}
)
:num_batches 16
:inner_rows_total 2000306.4
}
}
:aggstrategy 1
:aggsplit 6
:numGroups 5
:transitionSpace 0
:groupClause {
  {SORTGROUPCLAUSE
    :tSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
}
)
:qual <>
}
}
:num_workers 2
}
}
:aggstrategy 1
:aggsplit 9
:numGroups 5
:transitionSpace 0
:groupClause {
  {SORTGROUPCLAUSE
    :tSortGroupRef 1
    :eqop 98
    :sortop 664
    :reverse_sort false
    :nulls_first false
    :hashable true
  }
}
}
:qual <>
}
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded true
:jitFlags 25
:planTree
  {AGG
    :plan.disabled_nodes 0
    :plan.startup_cost 194594.51589866364
    :plan.total_cost 194617.5859964136
    :plan.plan_rows 5
    :plan.plan_width 17
    :plan.parallel_aware false
    :plan.parallel_safe false
    :plan.async_capable false
  }
}

```

```

:plan.plan_node_id 0
:plan.targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno -2
      :varattno 1
      :vartype 25
      :vartypmod -1
      :varcollid 100
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 0
      :varattnosyn 0
      :location -1
    }
    :resno 1
    :resname o_orderpriority
    :ressortgroupref 1
    :resorigtbl 16973
    :resorigcol 6
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {AGGREF
      :aggfnoid 2803
      :aggtype 20
      :aggcollid 0
      :inputcollid 0
      :aggtranstype 20
      :aggargtypes <>
      :aggdirectargs <>
      :args (
        {TARGETENTRY
          :expr
          {VAR
            :varno -2
            :varattno 2
            :vartype 20
            :vartypmod -1
            :varcollid 0
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 0
            :varattnosyn 0
            :location -1
          }
          :resno 1
          :resname <>
          :ressortgroupref 0
          :resorigtbl 0
          :resorigcol 0
          :resjunk false
        }
      )
      :aggorder <>
      :aggdistinct <>
      :aggfilter <>
      :aggstar true
      :aggvariadic false
      :aggkind n
      :aggpresorted false
      :agglevelsup 0
      :aggssplit 9
      :aggno 0
      :aggtransno 0
      :location 32
    }
    :resno 2
    :resname order_count
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:plan.qual <>
:plan.lefttree
{GATHERMERGE
  :plan.disabled_nodes 0
  :plan.startup_cost 194594.51589866364
  :plan.total_cost 194617.47599641362
  :plan.plan_rows 12
  :plan.plan_width 17
  :plan.parallel_aware false
  :plan.parallel_safe false
  :plan.async_capable false
  :plan.plan_node_id 1
  :plan.targetlist (
    {TARGETENTRY
      :expr
      {VAR
        :varno -2
        :varattno 1
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 0
        :varattnosyn 0
        :location -1
      }
      :resno 1
      :resname <>
      :ressortgroupref 1
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    }
    {TARGETENTRY
      :expr
      {VAR
        :varno -2
        :varattno 2
        :vartype 20
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 0
        :varattnosyn 0
        :location -1
      }
      :resno 2
      :resname <>
      :ressortgroupref 0
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    }
  )
  :plan.qual <>
:plan.lefttree
{AGG
  :plan.disabled_nodes 0
  :plan.startup_cost 193594.49212422612
  :plan.total_cost 193616.06712422607
  :plan.plan_rows 5
  :plan.plan_width 17
  :plan.parallel_aware false
  :plan.parallel_safe true
  :plan.async_capable false

```

```

:plan.plan_node_id 2
:plan.targetlist (
  {TARGETENTRY
    :expr
      {VAR
        :varno -2
        :varattno 1
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 0
        :varattnosyn 0
        :location -1
      }
    :resno 1
    :resname <>
    :ressortgroupref 1
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
  {TARGETENTRY
    :expr
      {AGGREF
        :aggfnoid 2803
        :aggtype 20
        :aggcollid 0
        :inputcollid 0
        :aggtranstype 20
        :aggargtypes <>
        :aggdirectargs <>
        :args <>
        :aggorder <>
        :aggdistinct <>
        :aggfilter <>
        :aggstar true
        :aggvariadic false
        :aggkind n
        :aggpresorted false
        :agglevelsup 0
        :aggsplit 6
        :aggno 0
        :aggtransno 0
        :location 32
      }
    :resno 2
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:plan.qual <>
:plan.lefttree
  {SORT
    :plan.disabled_nodes 0
    :plan.startup_cost 193594.49212422612
    :plan.total_cost 193601.6671242261
    :plan.plan_rows 2870
    :plan.plan_width 9
    :plan.parallel_aware false
    :plan.parallel_safe true
    :plan.async_capable false
    :plan.plan_node_id 3
    :plan.targetlist (
      {TARGETENTRY
        :expr
          {VAR
            :varno -2
            :varattno 1
            :vartype 25
            :vartypmod -1
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 4
            :varattnosyn 6
            :location -1
          }
        :resno 1
        :resname <>
        :ressortgroupref 1
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
      }
    )
  }
:plan.qual <>
:plan.lefttree
  {HASHJOIN
    :join.plan.disabled_nodes 0
    :join.plan.startup_cost 154455.46666666667
    :join.plan.total_cost 193429.6560416667
    :join.plan.plan_rows 2870
    :join.plan.plan_width 9
    :join.plan.parallel_aware true
    :join.plan.parallel_safe true
    :join.plan.async_capable false
    :join.plan.plan_node_id 4
    :join.plan.targetlist (
      {TARGETENTRY
        :expr
          {VAR
            :varno -2
            :varattno 1
            :vartype 25
            :vartypmod -1
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 4
            :varattnosyn 6
            :location -1
          }
        :resno 1
        :resname <>
        :ressortgroupref 1
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
      }
    )
  }
)
:join.plan.qual <>
:join.plan.lefttree
  {HASHJOIN
    :join.plan.disabled_nodes 0
    :join.plan.startup_cost 4806.4125
    :join.plan.total_cost 40362.284374999996
    :join.plan.plan_rows 10210
    :join.plan.plan_width 13
    :join.plan.parallel_aware true
    :join.plan.parallel_safe true
    :join.plan.async_capable false
    :join.plan.plan_node_id 5
    :join.plan.targetlist (
      {TARGETENTRY
        :expr
          {VAR
            :varno -2
            :varattno 6
            :vartype 25
            :vartypmod -1

```

```

:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:join.plan.qual <>
:join.plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 35383
:scan.plan.plan_rows 22585
:scan.plan.plan_width 17
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 6
:scan.plan.targetlist {
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location -1
}
:resno 3
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 4
:location -1
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)

```

```

        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 5
        :location -1
    }
    :resno 5
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 4
    :varattno 6
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 6
    :location -1
    }
    :resno 6
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 4
    :varattno 7
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 7
    :location -1
    }
    :resno 7
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 4
    :varattno 8
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 8
    :location -1
    }
    :resno 8
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    {TARGETENTRY
    :expr
    {VAR
    :varno 4
    :varattno 9
    :vartype 25
    :vartypmod -1
    :varcollid 100
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 9
    :location -1
    }
    :resno 9
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
    }
    }
    :scan.plan.qual (
    {OPEXPR
    :opno 1098
    :opfuncid 1090
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 4
    :varattno 5
    :vartype 1082
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 5
    :location -1
    }
    {CONST
    :consttype 1082
    :consttypmod -1
    :constcollid 0
    :constlen 4
    :constbyval true
    :constisnull false
    :location 109
    :constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
    }
    )
    :location 101
    }
    {OPEXPR
    :opno 2345
    :opfuncid 2338
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
    {VAR
    :varno 4
    :varattno 5

```

```

        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 5
        :location -1
    }
    {CONST
    :consttype 1114
    :consttypmod -1
    :constcollid 0
    :constlen 8
    :constbyval true
    :constisnull false
    :location -1
    :constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
    }
    }
    :location 142
    }
    }
    :scan.plan.lefttree <>
    :scan.plan.righttree <>
    :scan.plan.initPlan <>
    :scan.plan.extParam (b 0)
    :scan.plan.allParam (b 0)
    :scan.scanreliid 4
    }
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 4453.25
:plan.total_cost 4453.25
:plan.plan_rows 28253
:plan.plan_width 4
:plan.parallel_aware true
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 7
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 4453.25
:scan.plan.plan_rows 28253
:scan.plan.plan_width 4
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 8
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OPEXPR
:opno 674
:opfuncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
)
:location -1
}
)
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanreliid 5
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)

```

```

:plan.allParam (b 0)
:hashkeys (
  {VAR
    :varno -2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 1
    :location -1
  }
)
:skewTable 16561
:skewColumn 2
:skewInherit false
:rows_total 67807.2
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -2
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location -1
      }
      {VAR
        :varno -1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 5
        :varattnosyn 1
        :location -1
      }
    )
    :location -1
  }
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
  {VAR
    :varno -2
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 2
    :location -1
  }
)
}
:join.plan.righttree
{HASH
  :plan.disabled_nodes 0
  :plan.startup_cost 135974.79166666666
  :plan.total_cost 135974.79166666666
  :plan.plan_rows 833461
  :plan.plan_width 4
  :plan.parallel_aware true
  :plan.parallel_safe true
  :plan.async_capable false
  :plan.plan_node_id 9
  :plan.targetlist (
    {TARGETENTRY
      :expr
      {VAR
        :varno -2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location -1
      }
      :resno 1
      :resname <>
      :resortgroupref 0
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    }
  )
  :plan.qual <>
  :plan.lefttree
  {SESCAN
    :scan.plan.disabled_nodes 0
    :scan.plan.startup_cost 0
    :scan.plan.total_cost 135974.79166666666
    :scan.plan.plan_rows 833461
    :scan.plan.plan_width 4
    :scan.plan.parallel_aware true
    :scan.plan.parallel_safe true
    :scan.plan.async_capable false
    :scan.plan.plan_node_id 10
    :scan.plan.targetlist (
      {TARGETENTRY
        :expr
        {VAR
          :varno 3
          :varattno 1
          :vartype 23
          :vartypmod -1
          :varcollid 0
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 3
          :varattnosyn 1
          :location 289
        }
        :resno 1
      }
    )
  }
}

```

```

        :resname <>
        :ressortgroupref 0
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
    }
}
:scan.plan.qual (
{OEXPR
:opno 1895
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location 329
}
{VAR
:varno 3
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location 344
}
)
:location 342
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 3
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 289
}
)
)
:skewTable 16561
:skewColumn 1
:skewInherit false
:rows_total 2000306.4
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
{VAR
:varno -1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location 289
}
)
)
:location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)

```

```

        :nullsFirst ( false)
    }
    :plan.righttree <>
    :plan.initPlan <>
    :plan.extParam (b 0)
    :plan.allParam (b 0)
    :aggstrategy 1
    :aggsplit 6
    :numCols 1
    :grpColIdx ( 1)
    :grpOperators ( 98)
    :grpCollations ( 100)
    :numGroups 5
    :transitionSpace 0
    :aggParams (b)
    :groupingSets <>
    :chain <>
    }
    :plan.righttree <>
    :plan.initPlan <>
    :plan.extParam (b)
    :plan.allParam (b)
    :num_workers 2
    :rescan_param 0
    :numCols 1
    :sortColIdx ( 1)
    :sortOperators ( 664)
    :collations ( 100)
    :nullsFirst ( false)
    :initParam (b)
    }
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b)
:plan.allParam (b)
:aggstrategy 1
:aggsplit 9
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 5
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 16973
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:grouppr <>
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_lines" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname c
:colnames <>
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}

```

```

    }
}
:unprunableRelids (b 1 3 4 5)
:permInfos (
  {RTEPERMISSIONINFO
    :relid 16973
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 12 13)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16566
    :inh true
    :requiredPerms 2
    :checkAsUser 0
    :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16561
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 9 10 11 12 13 14 15 16)
    :insertedCols (b)
    :updatedCols (b)
  }
  {RTEPERMISSIONINFO
    :relid 16546
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:resultRelations <>
:appendRelations <>
:subplans <>
:rewindPlanIDs (b)
:rowMarks <>
:relationOids (o 16973 16566 16561 16546)
:invalItems <>
:paramExecTypes (o 0)
:utilityStmt <>
:stmt_location 0
:stmt_len 421
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL: (PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 25
:planTree
  {AGG
    :plan.disabled_nodes 0
    :plan.startup_cost 193594.49212422612
    :plan.total_cost 193616.06712422607
    :plan.plan_rows 5
    :plan.plan_width 17
    :plan.parallel_aware false
    :plan.parallel_safe true
    :plan.async_capable false
    :plan.plan_node_id 2
    :plan.targetlist (
      {TARGETENTRY
        :expr
          {VAR
            :varno -2
            :varattno 1
            :vartype 25
            :vartypmod -1
            :varcollid 100
            :varnullingrels (b)
            :varlevelsup 0
            :varreturningtype 0
            :varnosyn 0
            :varattnosyn 0
            :location -1
          }
        :resno 1
        :resname <>
        :ressortgroupref 1
        :resorigtbl 0
        :resorigcol 0
        :resjunk false
      }
    )
    {TARGETENTRY
      :expr
        {AGGREF
          :aggfnoid 2803
          :aggtype 20
          :aggcollid 0
          :inputcollid 0
          :aggtranstype 20
          :aggargtypes <>
          :aggdirectargs <>
          :args <>
          :aggorder <>
          :aggdistinct <>
          :aggfilter <>
          :aggstar true
          :aggvariadic false
          :aggkind n
          :aggpresorted false
          :agglevelsup 0
          :aggsplit 6
          :aggno 0
          :aggtransno 0
          :location -1
        }
      :resno 2
      :resname <>
      :ressortgroupref 0
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    )
  }
:plan.qual <>
:plan.lefttree
  {SORT
    :plan.disabled_nodes 0
    :plan.startup_cost 193594.49212422612
    :plan.total_cost 193601.6671242261
    :plan.plan_rows 2870
    :plan.plan_width 9
    :plan.parallel_aware false
    :plan.parallel_safe true
    :plan.async_capable false
    :plan.plan_node_id 3
    :plan.targetlist (
      {TARGETENTRY
        :expr
          {VAR

```

```

:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 154455.46666666667
:join.plan.total_cost 193429.6560416667
:join.plan.plan_rows 2870
:join.plan.plan_width 9
:join.plan.parallel_aware true
:join.plan.parallel_safe true
:join.plan.async_capable false
:join.plan.plan_node_id 4
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:join.plan.qual <>
:join.plan.lefttree
{HASHJOIN
:join.plan.disabled_nodes 0
:join.plan.startup_cost 4806.4125
:join.plan.total_cost 40362.284374999996
:join.plan.plan_rows 10210
:join.plan.plan_width 13
:join.plan.parallel_aware true
:join.plan.parallel_safe true
:join.plan.async_capable false
:join.plan.plan_node_id 5
:join.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:join.plan.qual <>
:join.plan.lefttree
{SESCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 35383
:scan.plan.plan_rows 22585
:scan.plan.plan_width 17
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 6
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)

```

```

}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location -1
}
:resno 3
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 4
:location -1
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
:resno 5
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 6
:location -1
}
:resno 6
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 7
:location -1
}
:resno 7
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 8
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 8
:location -1
}
}

```

```

:resno 8
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 9
:location -1
}
:resno 9
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:scan.plan.qual (
{OEXPR
:opno 1098
:opfuncid 1098
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 -1 ]
}
)
:location -1
}
{OEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
)
:location -1
}
}
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 4
}
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 4453.25
:plan.total_cost 4453.25
:plan.plan_rows 28253
:plan.plan_width 4
:plan.parallel_aware true
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 7
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 4453.25

```

```

:scan.plan.plan_rows 28253
:scan.plan.plan_width 4
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 8
:scan.plan.targetlist (
  {TARGETENTRY
    :expr
    {VAR
      :varno 5
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 5
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:scan.plan.qual (
  {OPEXPR
    :opno 674
    :opfuncid 297
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 5
        :varattno 6
        :vartype 701
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 5
        :varattnosyn 6
        :location -1
      }
      {CONST
        :consttype 701
        :consttypmod -1
        :constcollid 0
        :constlen 8
        :constbyval true
        :constisnull false
        :location -1
        :constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
      }
    )
    :location -1
  }
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 5
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
  {VAR
    :varno -2
    :varattno 1
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 5
    :varattnosyn 1
    :location -1
  }
)
:skewTable 16561
:skewColumn 2
:skewInherit false
:rows_total 67807.2
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -2
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location -1
      }
      {VAR
        :varno -1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 5
        :varattnosyn 1
        :location -1
      }
    )
    :location -1
  }
)
}
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
  {VAR
    :varno -2
    :varattno 2

```

```

        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location -1
    }
}
}
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 135974.79166666666
:plan.total_cost 135974.79166666666
:plan.plan_rows 833461
:plan.plan_width 4
:plan.parallel_aware true
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 9
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 135974.79166666666
:scan.plan.plan_rows 833461
:scan.plan.plan_width 4
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 10
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 3
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OPEXPR
:opno 1095
:opfuncid 1087
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 3
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 12
:location -1
}
{VAR
:varno 3
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location -1
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 3
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
)
)

```

```

:skewTable 16561
:skewColumn 1
:skewinherit false
:rows_total 2000306.4
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.jointype 4
:join.inner_unique false
:join.joinqual <>
:hashclauses (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opreset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -2
:varattno 2
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
{VAR
:varno -1
:varattno 1
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
)
:location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
{VAR
:varno -2
:varattno 2
:varatype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:aggstrategy 1
:aggsplit 6
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 5
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:table (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname orders_secure
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 16973
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
:rtekind 9
:groupexprs <>
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesstatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true

```

```

:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 16561
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesampl <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname c
:colnames <>
}
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
:rtekind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesampl <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
:unprunableRelids (b 1 3 4 5)
:permInfos (
{RTEPERMISSIONINFO
:relid 16973
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
:resultRelations <>
:appendRelations <>
:subplans <>
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalItems <>
:paramExecTypes (o 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

```

WARNING: Log 4: At the Executor starting point we have proposed Plan by Planner:

```

DETAIL:  {PLANNEDSTMT
:commandType 1
:queryId 0
:planId 0
:hasReturning false
:hasModifyingCTE false
:canSetTag true
:transientPlan false
:dependsOnRole false
:parallelModeNeeded false
:jitFlags 25
:planTree
{AGG
:plan.disabled_nodes 0
:plan.startup_cost 193594.49212422612
:plan.total_cost 193616.06712422607
:plan.plan_rows 5
:plan.plan_width 17
:plan.parallel_aware false
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 2
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 0
:varattnosyn 0
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 1
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY

```

```

:expr
{
  AGGREF
  :aggfnoid 2883
  :aggtype 28
  :aggcollid 0
  :inputcollid 0
  :aggtranstype 28
  :aggargtypes <>
  :aggdirectargs <>
  :args <>
  :aggorder <>
  :aggdistinct <>
  :aggfilter <>
  :aggstar true
  :aggvariadic false
  :aggkind n
  :aggresorted false
  :agglevelsup 0
  :aggssplit 6
  :aggno 0
  :aggtransno 0
  :location -1
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{
  SORT
  :plan.disabled_nodes 0
  :plan.startup_cost 193594.49212422612
  :plan.total_cost 193601.6671242261
  :plan.plan_rows 2870
  :plan.plan_width 9
  :plan.parallel_aware false
  :plan.parallel_safe true
  :plan.async_capable false
  :plan.plan_node_id 3
  :plan.targetlist (
    {TARGETENTRY
      :expr
      {
        VAR
        :varno -2
        :varattno 1
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 6
        :location -1
      }
      :resno 1
      :resname <>
      :ressortgroupref 1
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    }
  )
}
:plan.qual <>
:plan.lefttree
{
  HASHJOIN
  :join.plan.disabled_nodes 0
  :join.plan.startup_cost 154455.46666666667
  :join.plan.total_cost 193429.6560416667
  :join.plan.plan_rows 2870
  :join.plan.plan_width 9
  :join.plan.parallel_aware true
  :join.plan.parallel_safe true
  :join.plan.async_capable false
  :join.plan.plan_node_id 4
  :join.plan.targetlist (
    {TARGETENTRY
      :expr
      {
        VAR
        :varno -2
        :varattno 1
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 6
        :location -1
      }
      :resno 1
      :resname <>
      :ressortgroupref 1
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    }
  )
}
:join.plan.qual <>
:join.plan.lefttree
{
  HASHJOIN
  :join.plan.disabled_nodes 0
  :join.plan.startup_cost 4806.4125
  :join.plan.total_cost 48362.284374999996
  :join.plan.plan_rows 10210
  :join.plan.plan_width 13
  :join.plan.parallel_aware true
  :join.plan.parallel_safe true
  :join.plan.async_capable false
  :join.plan.plan_node_id 5
  :join.plan.targetlist (
    {TARGETENTRY
      :expr
      {
        VAR
        :varno -2
        :varattno 6
        :vartype 25
        :vartypmod -1
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 6
        :location -1
      }
      :resno 1
      :resname <>
      :ressortgroupref 0
      :resorigtbl 0
      :resorigcol 0
      :resjunk false
    }
  )
  {TARGETENTRY
    :expr
    {
      VAR
      :varno -2
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
    }
  }
}

```

```

        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 1
        :location -1
    }
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:join.plan.qual <>
:join.plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 35383
:scan.plan.plan_rows 22585
:scan.plan.plan_width 17
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 6
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 2
:location -1
}
}
:resno 2
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 3
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 3
:location -1
}
}
:resno 3
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 4
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 4
:location -1
}
}
:resno 4
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
}
:resno 5
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 6
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4

```

```

        :varattnosyn 6
        :location -1
    }
}
:resno 6
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 7
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 7
:location -1
}
}
:resno 7
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 8
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 8
:location -1
}
}
:resno 8
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 4
:varattno 9
:vartype 25
:vartypmod -1
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 9
:location -1
}
}
:resno 9
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
}
:scan.plan.qual (
{OPEXPR
:opno 1098
:opfuncid 1098
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
}
{CONST
:consttype 1082
:consttypmod -1
:constcollid 0
:constlen 4
:constbyval true
:constisnull false
:location -1
:constvalue 4 [ 113 -9 -1 -1 -1 -1 -1 ]
}
}
:location -1
}
{OPEXPR
:opno 2345
:opfuncid 2338
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 4
:varattno 5
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 5
:location -1
}
}
{CONST
:consttype 1114
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 32 105 23 -25 90 -1 -1 ]
}
}
:location -1
}
)
:scan.plan.lefttree <>

```

```

:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 4
}
:join.plan.righttree
{HASH
:plan.disabled_nodes 0
:plan.startup_cost 4463.25
:plan.total_cost 4463.25
:plan.plan_rows 28253
:plan.plan_width 4
:plan.parallel_aware true
:plan.parallel_safe true
:plan.async_capable false
:plan.plan_node_id 7
:plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:plan.qual <>
:plan.lefttree
{SEQSCAN
:scan.plan.disabled_nodes 0
:scan.plan.startup_cost 0
:scan.plan.total_cost 4463.25
:scan.plan.plan_rows 28253
:scan.plan.plan_width 4
:scan.plan.parallel_aware true
:scan.plan.parallel_safe true
:scan.plan.async_capable false
:scan.plan.plan_node_id 8
:scan.plan.targetlist (
{TARGETENTRY
:expr
{VAR
:varno 5
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
:resno 1
:resname <>
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:scan.plan.qual (
{OPEXPR
:opno 674
:opfncid 297
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno 5
:varattno 6
:vartype 701
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 6
:location -1
}
{CONST
:consttype 701
:consttypmod -1
:constcollid 0
:constlen 8
:constbyval true
:constisnull false
:location -1
:constvalue 8 [ 0 0 0 0 0 -120 -77 64 ]
}
)
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 5
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 5
:varattnosyn 1
:location -1
}
)
:skewTable 16561
:skewColumn 2
:skewInherit false
:rows_total 67807.2
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.jointype 4

```

```

:join.inner_unique false
:join.joinqual <>
:hashclauses (
  {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno -2
        :varattno 2
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 4
        :varattnosyn 2
        :location -1
      }
      {VAR
        :varno -1
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 5
        :varattnosyn 1
        :location -1
      }
    )
    :location -1
  }
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
  {VAR
    :varno -2
    :varattno 2
    :vartype 23
    :vartypmod -1
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 4
    :varattnosyn 2
    :location -1
  }
)
:join.plan.righttree
{HASH
  :plan.disabled_nodes 0
  :plan.startup_cost 135974.79166666666
  :plan.total_cost 135974.79166666666
  :plan.plan_rows 833461
  :plan.plan_width 4
  :plan.parallel_aware true
  :plan.parallel_safe true
  :plan.async_capable false
  :plan.plan_node_id 9
  :plan.targetlist (
    {TARGETENTRY
      :expr
      {VAR
        :varno -2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location -1
      }
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:plan.qual <>
:plan.lefttree
{SESCAN
  :scan.plan.disabled_nodes 0
  :scan.plan.startup_cost 0
  :scan.plan.total_cost 135974.79166666666
  :scan.plan.plan_rows 833461
  :scan.plan.plan_width 4
  :scan.plan.parallel_aware true
  :scan.plan.parallel_safe true
  :scan.plan.async_capable false
  :scan.plan.plan_node_id 10
  :scan.plan.targetlist (
    {TARGETENTRY
      :expr
      {VAR
        :varno 3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location -1
      }
    }
    :resno 1
    :resname <>
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:scan.plan.qual (
  {OPEXPR
    :opno 1095
    :opfuncid 1087
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
        :varno 3
        :varattno 12
        :vartype 1082
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
      }
    )
  }
)

```

```

:varnosyn 3
:varattnosyn 12
:location -1
}
{VAR
:varno 3
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 13
:location -1
}
}
:location -1
}
)
:scan.plan.lefttree <>
:scan.plan.righttree <>
:scan.plan.initPlan <>
:scan.plan.extParam (b 0)
:scan.plan.allParam (b 0)
:scan.scanrelid 3
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:hashkeys (
{VAR
:varno -2
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
)
:skewTable 16561
:skewColumn 1
:skewInherit false
:rows_total 2000306.4
}
:join.plan.initPlan <>
:join.plan.extParam (b 0)
:join.plan.allParam (b 0)
:join.jointype 4
:join.inner.unique false
:join.joinqual <>
:hashclauses (
{OEXPR
:opno 96
:opfuncid 65
:opresulttype 16
:opretset false
:opcollid 0
:inputcollid 0
:args (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
{VAR
:varno -1
:varattno 1
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 3
:varattnosyn 1
:location -1
}
)
:location -1
}
)
:hashoperators (o 96)
:hashcollations (o 0)
:hashkeys (
{VAR
:varno -2
:varattno 2
:vartype 23
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 4
:varattnosyn 1
:location -1
}
)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:numCols 1
:sortColIdx ( 1)
:sortOperators ( 664)
:collations ( 100)
:nullsFirst ( false)
}
:plan.righttree <>
:plan.initPlan <>
:plan.extParam (b 0)
:plan.allParam (b 0)
:aggstrategy 1
:aggssplit 6
:numCols 1
:grpColIdx ( 1)
:grpOperators ( 98)
:grpCollations ( 100)
:numGroups 5
:transitionSpace 0
:aggParams (b)
:groupingSets <>
:chain <>
}
:partPruneInfos <>
:table (
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname orders_secure

```

```

        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
    }
}
:rtckind 1
:subquery <>
:security_barrier false
:relid 16973
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname *GROUP*
:colnames ("o_orderpriority")
}
}
:rtckind 9
:groupexprs <>
:lateral false
:inFromCl false
:securityQuals <>
}
{RANGETBLENTY
:alias <>
:eref
{ALIAS
:aliasname lineitem
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenum" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
}
:rtckind 0
:relid 16566
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname o
:colnames <>
}
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
}
:rtckind 0
:relid 16561
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBLENTY
:alias
{ALIAS
:aliasname c
:colnames <>
}
}
:eref
{ALIAS
:aliasname c
:colnames ("c_custkey" "c_name" "c_address" "c_nationkey" "c_phone" "c_acctbal" "c_mktsegment" "c_comment")
}
}
:rtckind 0
:relid 16546
:inh false
:relkind r
:rellockmode 1
:perminfoindex 4
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:unprunableRelids (b 1 3 4 5)
:permInfos (
{RTEPERMISSIONINFO
:relid 16973
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 12 13)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16566
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16561
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 16546
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
}
)
:resultRelations <>
:appendRelations <>
:subplans <>
:rewindPlanIDs (b)
:rowMarks <>
:relationOids <>
:invalidItems <>
:paramExecTypes (o 0)
:utilityStmt <>
:stmt_location -1
:stmt_len -1
}

```